

Predicting the 2026 FIFA World Cup: A Machine Learning Approach with Monte Carlo Simulation

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Abstract

We present a machine learning pipeline for predicting international football match outcomes and simulating the 2026 FIFA World Cup. Using 21,371 matches from 2004–2026, we engineer 31 features spanning Elo ratings, FIFA rankings, Transfermarkt market valuations, player injuries, team form, head-to-head records, and World Cup history. We train and evaluate seven classifiers—Logistic Regression, KNN, Random Forest, XGBoost, HistGradientBoosting, SVM, and a Stacking Ensemble—on a three-class prediction task (Home Win, Draw, Away Win). After Optuna hyperparameter tuning and class imbalance handling, the best model (XGBoost with balanced class weights) achieves a macro F1 of 0.517 and an AUC-ROC of 0.710 on the validation set. Draw prediction remains the primary challenge, with the balanced model achieving 25.7% draw recall versus near-zero for unbalanced models. We validate against the 2022 World Cup (45.3% match accuracy) and run a 10,000-iteration Monte Carlo simulation of the full 2026 tournament with 48 teams across 104 matches.

1 Introduction

The 2026 FIFA World Cup represents the largest expansion in tournament history, growing from 32 to 48 teams across 104 matches hosted in the United States, Canada, and Mexico. This expansion introduces unprecedented prediction challenges: more teams from diverse confederations, a new 12-group format with third-place advancement rules, and a deeper knockout bracket.

International football prediction is inherently difficult. Unlike club football, national teams play infrequently (10–15 matches per year), rosters change between matches, and the competitive landscape shifts over multi-year cycles. The three-class prediction problem (Home Win, Draw, Away Win) is further complicated by the draw class, which occurs in approximately 23% of matches and lacks clear feature separation from narrow wins or losses.

Our approach combines traditional Elo-based features with modern market value data from Transfermarkt, injury information, and World Cup historical performance metrics. We explicitly avoid EA Sports FC video game ratings, which are

subjective editorial scores covering only a fraction of international football.

2 Background

2.1 Elo Ratings

The Elo rating system, adapted from chess, assigns each team a numerical strength rating updated after every match. The expected outcome is computed as:

$$W_e = \frac{1}{10^{-\Delta r/400} + 1} \quad (1)$$

where Δr is the rating difference between teams. After a match, ratings are updated: $R_n = R_o + K \times (W - W_e)$, where K varies by tournament importance (60 for World Cup finals, 20 for friendlies).

2.2 Gradient Boosting

XGBoost and HistGradientBoosting build ensembles of decision trees sequentially, with each tree correcting the errors of its predecessors. For multiclass classification, the model outputs class probabilities via softmax, optimizing multinomial log loss.

2.3 Monte Carlo Simulation

We simulate the full tournament 10,000 times, sampling match outcomes from predicted class probabilities at each stage. Draws in knockout rounds are resolved via a penalty shootout model calibrated from historical World Cup data.

3 Related Work

Prior work on World Cup prediction spans statistical models [1], machine learning approaches [2], and hybrid Elo-based systems. Groll et al. used random forests on FIFA ranking features for the 2018 World Cup. Hubacek et al. combined Elo ratings with bookmaker odds. Our work differs in three key ways: (1) we use Transfermarkt market values instead of EA FC ratings, (2) we incorporate injury data weighted by

player market value, and (3) we address the 48-team format with third-place advancement rules.

4 Dataset

4.1 Data Sources

We compile data from seven sources:

1. **International match results** (Kaggle): 21,371 matches from 2004–2026 with scores, tournaments, and venues.
2. **Elo ratings** (elratings.net): Live per-match Elo ratings for all 48 qualified teams.
3. **FIFA World Rankings**: Official ranking points and positions.
4. **Transfermarkt**: Player market values, squad compositions, and manager records for 30,000+ players.
5. **Injury histories**: 92,671 player injury records for historical training; live scraping for 2026 predictions.
6. **Historical World Cup data**: Match-level data for all World Cups from 1930–2022.
7. **2026 fixtures**: Official 72 group stage matches and knockout bracket from the FIFA draw.

4.2 Feature Engineering

All team-specific features are computed as deltas (home minus away) to capture relative strength. Our 31 features span six categories:

Rankings (4): Elo difference, Elo momentum (5-game change), FIFA rank difference, FIFA points difference.

Squad Strength (7): Total, top-11, positional (attack/midfield/defense), star player, and squad depth market value deltas, all log-transformed.

Form (5): Win rate, weighted form (exponential decay $\times$ importance), goal difference, and days of rest for each team.

Head-to-Head (3): Historical win rate, draw rate, and matches played between the two teams.

World Cup History (4): Appearances, knockout advancement rate, best finish (encoded 1–7), and goals per game.

Context (8): Match importance, neutral venue, tournament win rate, confederation overlap, continent advantage, injury burden, injury count, and star injury flag.

4.3 Target Variable and Class Distribution

The target is encoded as: 0 = Home Win (47.8%), 1 = Draw (23.4%), 2 = Away Win (28.8%). This class imbalance, particularly the underrepresented draw class, is a central challenge.

4.4 Chronological Splits

- **Train**: 2004-01-01 to 2019-12-31 (15,495 matches)
- **Validation**: 2020-01-01 to 2022-11-19 (2,324 matches)
- **Test**: 2022-11-20 onward (3,552 matches, including 2022 World Cup)

5 Methodology

5.1 Models

We train seven classifiers:

1. **Logistic Regression**: Multinomial baseline with StandardScaler.
2. **KNN**: Tested $k \in \{5, 10, 20, 50\}$; best at $k = 10$.
3. **Random Forest**: 500 trees with median imputation.
4. **XGBoost**: Histogram-based gradient boosting with native NaN handling.
5. **HistGradientBoosting**: Sklearn’s native histogram-based GBM.
6. **SVM (RBF)**: Probability-calibrated with StandardScaler.
7. **Stacking Ensemble**: XGBoost + HistGBM + RF + LR with LR meta-learner.

5.2 Hyperparameter Tuning

We use Optuna with 50 trials each for XGBoost and HistGBM, optimizing macro F1 on the validation set. Key tuned parameters include learning rate, max depth, regularization, and sub-sampling.

5.3 Class Imbalance Strategies

We compare: (a) no handling, (b) balanced class weights via sample weighting, (c) SMOTE oversampling of the training set, and (d) a two-stage approach (win/not-win binary classifier, then draw/loss classifier).

6 Results

6.1 Validation Set Performance

The key finding is the accuracy–F1 trade-off: models with highest accuracy (Stacking at 61.2%) achieve this by predicting almost no draws. Class-balanced models sacrifice 3–4% accuracy but achieve 5–10 $\times$ higher draw recall and meaningfully better macro F1.

Model	Acc	F1	LogLoss	D-Recall
XGB Balanced	0.574	0.517	1.003	25.7%
HistGBM Balanced	0.546	0.509	0.919	30.9%
XGB Tuned	0.590	0.505	1.006	17.7%
HistGBM Tuned	0.605	0.503	0.891	12.9%
Stacking	0.612	0.465	0.860	3.0%
Log. Regression	0.609	0.456	0.863	1.9%

Table 1: Validation set metrics. Balanced class weights significantly improve draw detection (D-Recall) at a modest accuracy cost.

6.2 Test Set (2022 World Cup Holdout)

On the test set (post-November 2022), HistGBM Balanced achieves the best macro F1 of 0.513 with log loss 0.918. The model generalizes well, with metrics consistent between validation and test sets.

6.3 2022 World Cup Analysis

Predicting all 64 matches of the 2022 World Cup, the balanced XGBoost achieves 45.3% accuracy (29/64 correct). The model correctly predicts most results involving strong favorites (Qatar losses, Argentina wins) but struggles with upsets. Notably, the model assigned Argentina vs. Saudi Arabia a 97% home win probability—one of football’s greatest upsets.

6.4 SHAP Analysis

SHAP feature importance reveals Elo difference as the dominant predictor, followed by FIFA ranking difference and form metrics. The ablation study shows that removing the bottom 30% of features (9 of 31) slightly improves performance (F1: 0.477 $\rightarrow$ 0.492), suggesting some features add noise.

6.5 Time-Series Cross-Validation

Expanding window CV across tournament cycles (2010, 2014, 2018, 2022) shows stable and improving performance as training data grows (F1: 0.476 $\rightarrow$ 0.508), confirming the model benefits from more historical data.

7 2026 World Cup Predictions

Using 10,000 Monte Carlo iterations with the balanced XGBoost model, we simulate the full 48-team tournament. The simulation samples outcomes from predicted probabilities, uses Poisson goal models, and resolves knockout draws via a penalty model calibrated from historical World Cup data.

Top predicted winners include Brazil (6.8%), Argentina (6.5%), and Spain (3.9%). The low individual win probabilities reflect the expanded 48-team format—even the strongest teams must win 7 consecutive knockout matches. Group stage

predictions show most groups are competitive, with third-place advancement adding uncertainty.

8 Real-World Impact

8.1 Applications

Match outcome probabilities have applications in sports analytics, media predictions, fan engagement platforms, and betting markets. The Monte Carlo framework provides not just point predictions but full probability distributions over tournament outcomes.

8.2 Ethical Considerations

Sports prediction models raise concerns about gambling addiction and market manipulation. We emphasize that our model’s 45% match accuracy and the inherent unpredictability of football (upsets are part of the sport’s appeal) should caution against over-reliance on predictions for betting purposes.

9 Conclusion and Future Work

We present a comprehensive pipeline for international football prediction, achieving competitive performance on a challenging three-class problem. Key findings include:

- Class balancing via sample weights provides the best accuracy–F1 trade-off
- Elo ratings are the strongest single predictor category
- Draw prediction remains fundamentally difficult ($\sim$ 26% recall at best)
- Feature ablation suggests 22 features suffice

Future improvements could include: real-time injury integration from multiple sources, tactical/formation features, deep learning on sequential match data, and ensemble methods specifically targeting the draw class.

References

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